



A distributed flocking control strategy for UAV groups

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ABSTRACT

To ensure the information security of UAV groups, maximize the detection range of the UAV group and minimize the communication range, this paper improves Olfati-Saber's flocking algorithm using virtual leader and integrates the multiple UAVs with real environment. The improved swarm control algorithm can help to make all agents accurately track the speed of virtual leaders. Then, we introduce the conception of virtual communication circle to control the communication power of each UAV, and the moving function of the target position is improved to meet the limited conditions of given communication distance, to ensure the non-collision and stable communication of the UAV during the moving process. All UAVs in the group system will keep converging, avoiding collision and the speed is the same as that of the virtual pilot. Therefore, multiple UAVs can track the virtual pilot for cluster flight and a distributed cooperative control strategy can be acquired. The simulation results show that the strategy proposed in this paper makes a better communication performance among UAVs in real time, and the UAV groups achieve the information more timely, which is proved to be an effective way to ensure the security communication among multiple UAVs.

1. Introduction

Flocking UAV (unmanned aerial vehicle) is a kind of population system composed of a group of aerial robots. Compared with the traditional multi UAV system, it has the unique advantages of robustness, scalability, flexibility and economy, and a wide range of application prospects [1,2]. On one hand, people can reveal the emergence of group intelligence through the study of group behavior, which is a self-organizing process. The characteristics of simple individuals' overall intelligent behavior through interaction or cooperation are flocking intelligence which can be applied to optimization algorithms, such as genetic optimization algorithm (GA), ant colony optimization algorithm (ACO), particle swarm optimization algorithm (PSO), etc [3–5]; on the other hand, inspired by the behavior of swarm intelligence, people can learn from the wisdom of biology and apply swarm intelligence to the research of cooperative control of multi robot system. By designing a certain control algorithm or applying swarm intelligence optimization algorithm, the system as a whole can present the desired behaviors. The group UAV system adopts the distributed control strategy inspired by swarm intelligence, which has the characteristics of self-organization and distribution. These characteristics enable the group system to be insensitive to local individual failure and fault, which is not the feature of centralized control. In such system, an UAV is an agent with complete autonomy, which does not need any central control unit to coordinate many autonomous individuals [6]. Therefore, the individual

can respond to the dynamic environment quickly and flexibly without waiting for the command and information of the central control unit. Although this kind of individual behavior does not seem to have a global purpose, many individual behaviors can emerge a kind of expected group cooperation behavior when they interact and coordinate in the same environment. Therefore, it is one of the important means of UAV security control to analyze the social clustering and cooperation in biological group activities, simulate the activity process of UAV group and study the cooperation of multi-agents.

At present, quite a few universities and scientific research institutes have done some work on the cooperative control and security protection strategies of UAV groups: inspired by the information exchange between natural animal groups, Beard proposed the concept that multiple UAVs need to maintain the communicable range for collective operations on the basis of information perception between groups of creatures [7]. Bhattacharya proposed the concept that the UAV group should be able to maintain the communicable range for collective operations. The communication problem of UAV is solved by differential symmetry theory [8]. Shen et al. have studied the control search theory of multiple UAVs and a state graph of “rate of return” is proposed to search for multiple stationary targets as a theoretical basis [9]. Yang expounded the influence of the orderly behavior of ant colony foraging process, studied the self-organization of multi UAV tasks, and proposed an algorithm to improve the efficiency of UAV task execution and to reduce the overall channel interference during

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communication [10]. Zhang proposed an algorithm to attract and repel swarms by considering the range of agent communication [11]. Through the analysis of the research status in the field of group UAV cooperative control at home and abroad, we can see that the particularity of the wireless self-organizing network used in the communication of multiple UAVs causes that the combination of the communication security factors among UAVs and the UAV cooperative control method in the direction of cybernetics is a very comprehensive field. Moreover, it has a certain degree of complexity, and there are few researches about their combination.

In this paper, the aggregation problem of group UAVs is studied firstly. The possible multi UAV security communication problem is proposed by the study of the relationship between multi UAV communication and network topology, group formation size, etc. To solve this problem, a distributed cooperative control method is put forward by improving the intelligent algorithm in swarm intelligence theory. Then, through the combination of the new virtual communication ring, the improved target moving function and the improved swarm algorithm, this paper studies the communication factors affecting the UAV group dynamics to guarantees the information security of the UAV group, and achieves the goal of maximizing the detection range of the UAV group and minimizing the communication range of the UAV group. Finally, by the simulation and comparison of the algorithm in MATLAB and NS2 network environment, the convergence and effectiveness of the algorithm are proved.

The rest of this paper is as follows. In Section 2 we explain the principle model of UAV and related communication rules. Then the traditional flocking control algorithm is also introduced. In Section 3 we explain our improved scheme based on improved flocking algorithm with virtual leader and a complete UAV communication scheme is proposed. In Section 4 we provide the simulations and corresponding result analysis to verify our work. Lastly, we conclude this paper in Section 5.

2. Preliminary works

2.1. Flocking system model

In this paper, a three degree of freedom(DOF) autonomous UAV is considered. The motion model of UAV i is described by the following differential equations [12]:

$$\dot{x}_i = v_i \cos \chi_i \cos \gamma_i \quad (1)$$

$$\dot{y}_i = v_i \sin \chi_i \cos \gamma_i \quad (2)$$

$$\dot{z}_i = v_i \sin \gamma_i \quad (3)$$

$$\dot{V}_i = -g \sin \gamma_i + \frac{1}{m} T_i - D_i \quad (4)$$

$$\dot{\chi}_i = \frac{L_i \sin \phi_i}{m_i V_i \cos \gamma_i} \quad (5)$$

$$\dot{\gamma}_i = \frac{L_i \cos \phi_i - m_i g \cos \gamma_i}{m_i V_i} \quad (6)$$

The above describes the nonlinear relation between status vector $X = [x_i, y_i, z_i, v_i, \chi_i, \gamma_i]^T$ and control input vector $U = [\phi_i, L_i, T_i]^T$, where (x_i, y_i, z_i) is the inertia position of UAV, χ_i, γ_i, ϕ_i denote the track bearing, track inclination and roll angle; L_i and D_i are lift and drag force; m_i, v_i, g are the quality, line speed and gravitational acceleration. We let $q = (x_i, y_i, z_i)^T$, and the derivative to time is:

$$\dot{q} = \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{z}_i \end{bmatrix} = \begin{bmatrix} v_i \cos \gamma_i \cos \psi_i \\ v_i \cos \gamma_i \sin \psi_i \\ v_i \sin \gamma_i \end{bmatrix} \quad (7)$$

New control input is introduced as

$$U' = \begin{bmatrix} u'_1 \\ u'_2 \\ u'_3 \end{bmatrix} = \begin{bmatrix} (T_i - D_i)/m_i \\ L_i \cos \phi_i / m_i g \\ L_i \sin \phi_i / m_i g \end{bmatrix} \quad (8)$$

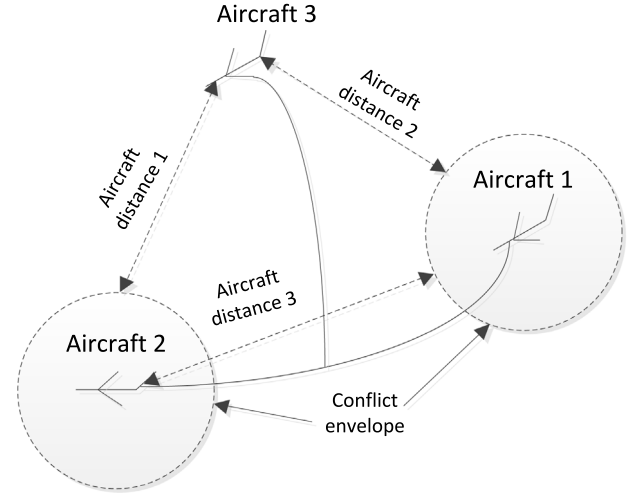


Fig. 1. Definition of the aircraft distance in the horizontal plane.

Then the nonlinear 3-DOF UAV model can be converted as a simple quadratic integral model

$$\ddot{q} = u_i \quad (9)$$

Thus, the swarm system constituted by 3-DOF UAVs can be described as the vector forms like

$$\dot{q} = p \quad (10)$$

$$\dot{p} = u \quad (11)$$

where $q = [q_1, q_2, \dots, q_n]^T$ is the location vector of system; $p = [p_1, p_2, \dots, p_n]^T$ is velocity vector; $u = [u_1, u_2, \dots, u_n]^T$ is the control input.

2.2. Wireless communication model

In the UAV system, the state information is transmitted by wireless communication between autonomous UAVs. The communication network model represents the relationship among the UAVs in the way of wireless communication [13]. If each UAV is regarded as a node in the algebraic graph, and the information relationship among UAVs is the edge of the graph, the multi UAV system can be seen as a graph, so the graph can be used to describe the information transfer relationship between UAVs (see Fig. 1).

The definition of communication topology is given to establish the communication network model of the system: the communication topology is represented by $G = \{V, E\}$, where the node set $V = \{1, 2, \dots, n\}$ represents n UAVs; the side set $E = \{(i, j) \in V \times V | i \sim j\}$ represents a set of communication links between UAVs. When there exists unidirectional communication between UAVs, the communication topology is a directed graph; when all of the communications among UAVs are bidirectional, the communication topology is undirected. In communication topology graph, the neighbor matrix is $A = [a_{ij}]$. If $(j, i) \in E$, $a_{ij} = 1$; otherwise, $a_{ij} = 0$. If there exists directed side $(j, i) \in E$, that is, the directed side from j to i , it means that j is transmitting state information to i . Then UAV j is the neighbor UAV of i . The neighbor UAV set of i can be computed as follows:

$$N_i = \{j \in V | a_{ij} = 1\} = \{j \in V | (j, i) \in E\} \quad (12)$$

2.3. Flocking control algorithm with virtual leader

Assuming the intelligence agent as node i , we use undirected edges to connect all the nodes in the neighbor region of i . Then, at time t , the network composed of these nodes $V = \{1, \dots, N\}$ and edge

$E(t) = \{(i, j) \in V \times V : i \sim j\}$ can be represented by an undirected graph $G(t)$. For certain intelligence agent, since some of the intelligence agents in its neighbor region range move to its neighbor region, or some of the intelligence agents out of the range of its neighbor region move to its neighbor region, the system has switchable topology [14]. The flocking control algorithm can be depicted as the method proposed in [15]:

$$u_i = - \sum_{j \in N_i(t)} \nabla q_i \psi_a(\|q_{ij} - q_{r_i}\|_\sigma) - \sum_{j \in N_i(t)} a_{ij}(t)(p_{ij} - p_{r_i}) + f_{r_i}(q_{r_i}, p_{r_i}) + c_1(q_{r_i} - q_i) + c_2(p_{r_i} - p_i) \quad (13)$$

where constant c_1 and c_2 represent the coefficient of leading feedback item; $\|z\|_\sigma$ has derivative everywhere except $\|z\| = 0$. The establishment of such norm is to construct smooth derivable artificial potential $\psi_a\|z\|_\sigma$, which is defined as

$$\begin{aligned} \psi_a(\|z\|_\sigma) &= \int_{\|d\|_\sigma}^{\|z\|_\sigma} a_{ij}(s)\phi(s - \|d\|_\sigma)d_s, \phi(\|z\|_\sigma) \\ &= \frac{1}{2}[(a+b) \frac{(\|z\|_\sigma + c)}{\sqrt{1 + ((\|z\|_\sigma) + c)^2}} + (a-b)] \end{aligned} \quad (14)$$

where a, b, c, d are normal numbers and $A(q) = (a_{ij}(q))$ is neighbor matrix.

Olfati Saber's flocking control algorithm adds the influence of virtual leaders, but he adopts single virtual leader. In this algorithm, the speed and location information of virtual leader can be sensed by all agents, and the algorithm can only make each agent match the virtual leader with a fixed speed. Therefore, the control input form of agent i in the system needs to be designed as follows:

$$u_i = f_i + h_i + \gamma_i, i = 1, 2, \dots, n \quad (15)$$

where f_i is gradient item of artificial potential energy function. All agents in the system keep connected and their positions gradually converge to a certain range. In the process of convergence, the agents can avoid collision with each other; h_i is speed control item, making the speed of all agents in the system gradually reach the same value; γ_i is leader information feedback item which aims to enable all agents in the system to track to the virtual leader.

3. Cooperative control strategy for safe communication distance of UAV

According to above analysis, we have certain knowledge of the possible security communication problems in multi UAV system, and how the communication affects the formation of multi UAV. Therefore, this paper improves the flocking algorithm with virtual leader in Olfati-Saber, so that each UAV node can accurately track the virtual leader with variable speed. Based on this method, the detection range of UAV group can be expanded and the communication range of UAV group can be minimized. Using the flocking algorithm based on the variable-speed virtual leader to control the UAV, the whole UAV group can reach the form of equal distance flocking formation, and finally form a distributed cooperative control strategy to solve the problem of multi UAV security communication.

3.1. Problem description

Assuming the moving model for each UAV in group system composed by n autonomous UAV is:

$$\begin{cases} \dot{q}_i = \dot{p}_i \\ \dot{p}_i = u_i \end{cases} \quad (16)$$

where q_i is the position of UAV i , p_i is the velocity of i , u_i is the control input of i .

The equation of virtual navigator motion is:

$$\begin{cases} \dot{q}_0 = p_0 \\ \dot{p}_0 = a_0 \end{cases} \quad (17)$$

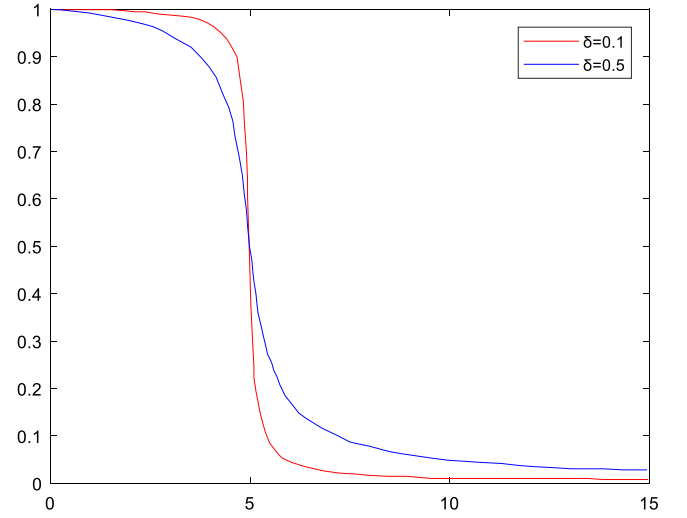


Fig. 2. Smoothing function.

If the communication radius of the UAVs in system are the same, the communication area for i is a circle which is also the neighborhood of i . Its adjacency set is

$$N_i = \{j \in V \mid a_{ij} = 1\} = \{j \mid \|q_i - q_j\| \leq r\} \quad (18)$$

Due to the limited communication area of the UAV, during the flying process, for a certain UAV, the adjacent UAV that has communication with it at the previous moment may move out of the communication area at the next moment. Therefore, the communication topological relationship between UAVs in the system is dynamic, and the communication topological diagram of the system is a dynamic diagram of continuous switching. We need to design a control strategy to keep the convergence, collision avoidance and speed of all UAVs in the system consistent, that is, the group UAV system achieves a group motion state, and the UAV group follows the virtual navigator.

We define smoothing function $\rho(x)$ is a scalar function changed from 0 to 1, as shown in Fig. 2.

$$\rho(x) = \left(\frac{r+\delta}{2r+\delta}\right) \left(1 - \frac{x-r}{\|x-r\|+\delta}\right) \quad (19)$$

The smoothing function is used to construct the smoothing matrix of communication topology $A = [a_{ij}]$, where the elements in matrix are defined as

$$a_{ij} = \begin{cases} \rho(\|q_{ij}\|) \in [0, 1] & j \neq i \\ 0 & j = i \end{cases} \quad (20)$$

When the distance between UAVs is greater than the communication radius r , there is no communication link relationship between UAVs. Since the communication radius of UAV is the same, the smooth adjacency matrix is symmetric matrix.

The key idea of the scheme is to use the improved virtual leader's flocking algorithm to control the UAV initially, so that the whole UAV group can reach the form of equal distance swarm formation. According to the GPS data, we know the absolute position of each UAV. To reduce the risk of information leakage, the direct idea is to minimize the communication range of UAVs outside the UAV group under the condition of meeting the communication needs. In order to make the individual UAV in the desired position and to meet the communication range conditions, the concept of virtual communication ring is proposed, and its establishment model is provided as the theoretical basis for the movement of each UAV. We draw on the idea in reference 1 to find the level node first. Then, according to this model, we calculate the communication radius of each UAV node, and define the radius of virtual communication circle as:

$$r_j = \min(r_{jk}) + r_{j-1} \quad (21)$$

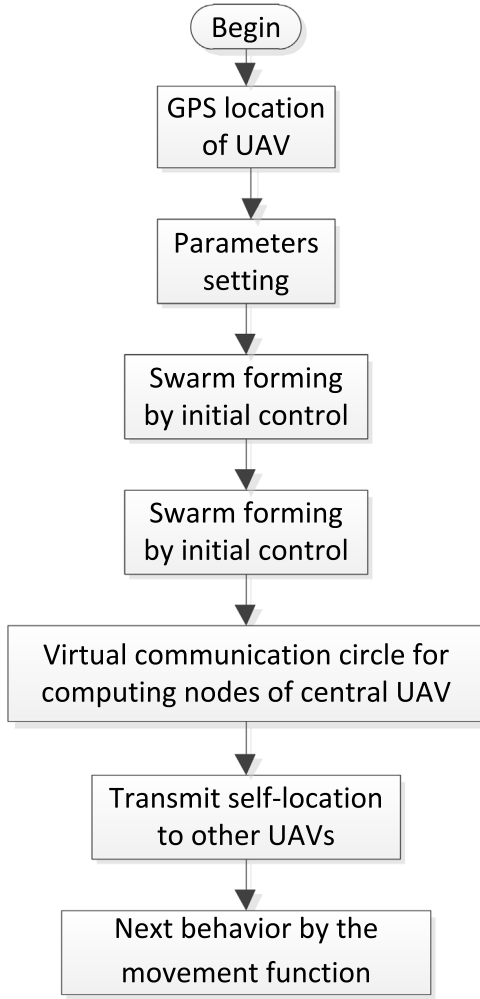


Fig. 3. Distributed control strategy among UAVs.

where r_{jk} represents the communication radius of k in the j_{th} layer.

3.2. The improvement of Olfati-Saber's algorithm

When the speed of virtual leader is constant, all agents with guidance information will converge to the speed of virtual leader [16]. The classical control algorithm only includes the feedback value of the virtual leader's speed, as is to know the speed information of the virtual leader. The convergence condition of the algorithm is that the neighborhood graph is always connected. Obviously, such condition cannot be achieved in reality. Without the location information of virtual leader, the flocking control algorithm cannot keep convergence in any switching topology. Therefore, an improved flocking control algorithm is proposed to make all agents track the speed of virtual leader accurately.

The difference between the location and speed of the agent and the location and speed of the virtual leader is set as $\tilde{q}_i = q_i - q_\gamma$ and $\tilde{p}_i = p_i - p_\gamma$, then

$$\dot{\tilde{q}}_i = \dot{\tilde{p}}_i \quad (22)$$

$$\dot{\tilde{q}}_i = \dot{\tilde{p}}_i = u_i - f_\gamma(q_\gamma, p_\gamma), \quad i = 1, 2, \dots, N \quad (23)$$

$$\tilde{V}_i = \sum_{j=1, j \neq i}^N \psi_a(\|\tilde{q}_{ij}\|_\sigma) = \sum_{j \neq -1, j \neq i} \psi_a(\|r\|_\sigma) + \sum_{j \in N_i(t)} \psi_a(\|\tilde{q}_{ij}\|_\sigma) \quad (24)$$

By control input and energy equation of tradition algorithm there are the following equations:

$$\ddot{u}_i = - \sum_{j \in N_i(t)} \nabla_{\tilde{q}_i} \psi_a(\|\tilde{q}_{ij}\|_\sigma) - \sum_{j \in N_i(t)} a_{ij}(t)(p_i - \tilde{p}_j) - [c_1 \tilde{q}_i + c_2 \tilde{p}_i] \quad (25)$$

$$Q = \frac{1}{2} \sum_{i=1}^N (U_i + \tilde{p}_i^T \tilde{p}_i) \quad (26)$$

where

$$U_i = \sum_{j=1, j \neq i}^N \psi_a(\|\tilde{q}_{ij}\|_\sigma) + c_1 \tilde{q}_i^T \tilde{q}_i + \tilde{V}_i + c_1 \tilde{q}_i^T \tilde{q}_i \quad (27)$$

Because of the symmetry of artificial potential function ψ_a and neighboring matrix $A(q)$,

$$\frac{1}{2} \sum_{i=1}^N \dot{U}_i = \sum_{i=1}^N (\tilde{p}_i^T \nabla_{\tilde{q}_i} \tilde{V}_i + c_1 \tilde{q}_i^T \dot{\tilde{q}}_i) \quad (28)$$

where $\dot{U}_i = dU_i/dt$. Therefore, we have

$$\dot{Q} = \frac{1}{2} \sum_{i=1}^N \dot{U}_i + \sum_{i=1}^N \tilde{p}_i^T \dot{\tilde{p}}_i = -\tilde{p}_i^T [(L(t) + c_2 I_N) \otimes I_N] \tilde{p} \quad (29)$$

Since $L(t)$ is positive semi-definite matrix, $\dot{Q} \leq 0$, that is, $Q(t)$ is a nonincreasing function. For any time t , $Q(t) \leq Q_0$. We also know that for any intelligence agent i , $c_1 \tilde{q}_i^T \tilde{q}_i \leq 2Q_0$, so the distance between any intelligence agent and virtual leader will be no more than $\sqrt{2Q_0/c_1}$. The improved flocking algorithm is

$$u_i = - \sum_{j \in N_i(t)} \nabla_{\tilde{q}_i} \psi_a(\|q_i - q_j\|_\sigma) - \sum_{j \in N_i(t)} a_{ij}(t)(p_i - p_j) + f_\gamma(q_\gamma, p_\gamma) + c_1(q_\gamma - q_i) + c_2(p_\gamma - p_i) \quad (30)$$

3.3. Equal distance controlling strategy among UAVs

In a multi UAV system, each UAV node has energy limitation, and the cruise and flight time cannot be too long. Therefore, the key issues to be considered in the mobile process lie in reaching the desired position constrained by the virtual communication loop, reducing energy consumption and increasing the endurance time. Based on the moving function of reference, a new UAV desired position acquisition model is designed, which is more suitable for multi UAV system with limited energy. Such function ensures that the UAV node can move to the desired position, meets the limited conditions of the communication ring, and ensures that the UAV does not collide in the process of moving. Therefore, we redefine the movement function and design a new UAV desired position acquisition model.

Supposing x_i, v_i, m_i, F_i are the vectors of position, velocity, material and control input. Since the local motion state information is updated by iteration computing to compute the objective position, X_i is set to denote the motion state of single node:

$$X_i(k) = \begin{bmatrix} x_i(k) \\ v_i(k) \end{bmatrix} \quad (31)$$

Such equation is discretized to acquire the iteration computation equation of node state:

$$X_i(k+1) = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_i(k) \\ v_i(k) \end{bmatrix} \quad (32)$$

$$X_i(k+1) = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_i(k) \\ v_i(k) \end{bmatrix} + \begin{bmatrix} t(T-t/2) \\ t \end{bmatrix} F_i(k)/m_i \quad (33)$$

where T is the time length of each step, $F_i(k)$ is the control input of node i at time k , and t is the action time of the control input. Through this equation, we can calculate the next moment's motion state from the current motion state information. According to the motion state update equation, in each slot T , the node will first make a uniform acceleration movement of T seconds. When the speed reaches the upper limit of a

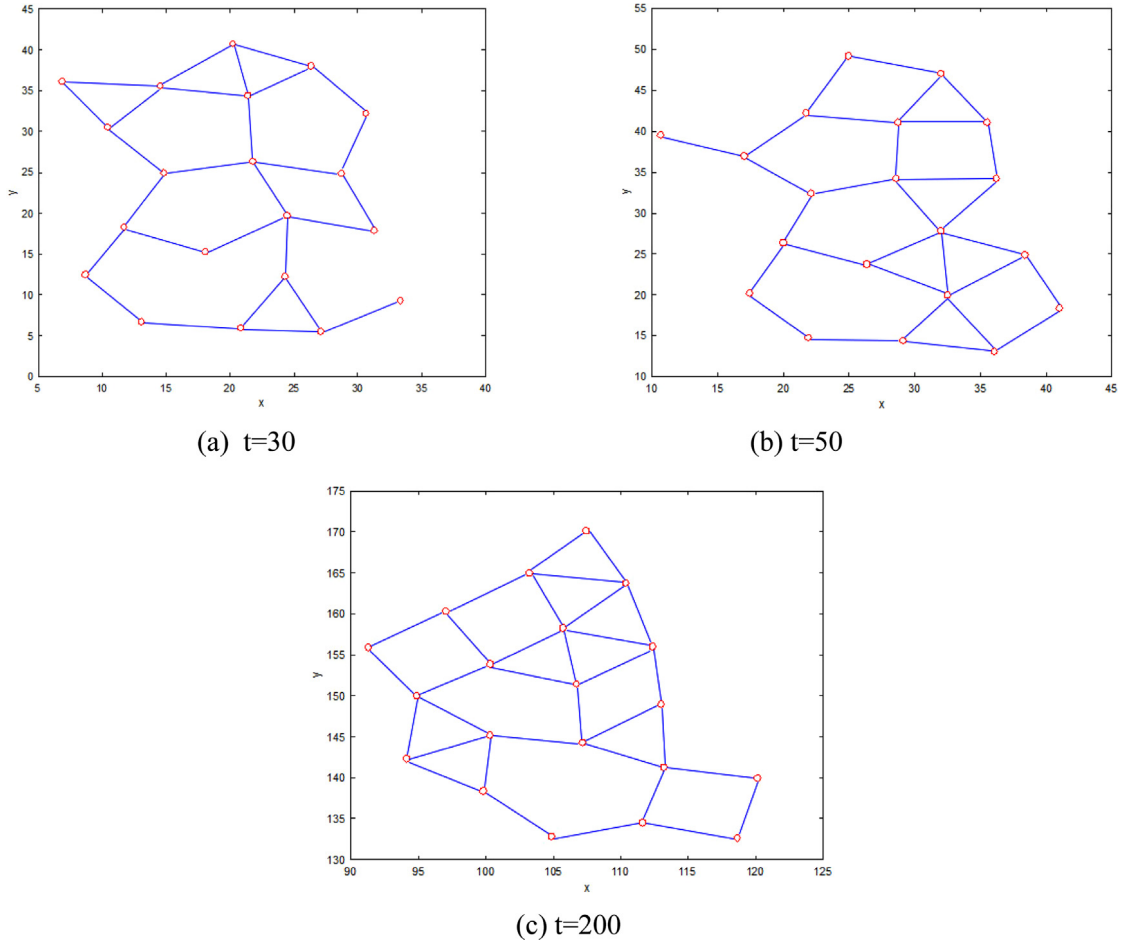


Fig. 4. Evolution figure of multi-agent system with virtual leader's changing rate.

node's speed, the node stops the acceleration movement and makes a uniform straight-line movement in the remaining time until the end of the movement process.

After the above parts of the description: collaborative control, virtual communication circle setting and target position moving function. We propose a distributed control method to solve the security problem among UAVs as the Fig. 3:

4. Simulations

4.1. Tests of improved flocking control algorithm

On MATLAB simulation platform, the multi-agent system composed of 20 agents in two-dimensional space is considered and its initialized distribution status is given: the location and direction angle of the intelligence agent are random generated in the range of $[0, 50] \times [0, 50]$ and $[-\pi, \pi]$, with velocity $v \in [1, 2]$; the initial location of virtual leader $q_v(0) = [20, 20]$ and speed $p_v(0) = 2\sqrt{2}$; Perception radius $R = 8$; expected distance $d = 7$; Feedback coefficient $c_1 = 0.4$ and $c_2 = 0.2$. Based on the classical flocking control model with virtual leader, this paper studies the influence of virtual leader on flocking control of multi-agent system at different speed. Fig. 4 shows the flocking evolution process of multi-agent system when the virtual leader rate is 5.0.

Firstly, based on the improved Olfati-Saber flocking control model, the impact of the improved virtual leader on the flocking control of multi-agent system is studied. As can be seen from this figure, when the virtual leader's speed is 5.0, the virtual leader quickly moves away

from other agents. In addition, the virtual leader starts to break away from the perception range of the common agent closest to it before and after the position shown in Fig. 4(b). The virtual leader no longer plays a guiding role for the common agent, that is, the control algorithm loses its role. Finally, under the influence of the first two items of the control algorithm, a stable α -lattice topology structure like Olfati-Saber model can be formed, as depicted in Fig. 4(c). It is verified that the state information of virtual leader plays an important role in regulating the flocking control of multi-agent system.

Fig. 5 is a graph of the relationship between the optimal speed of virtual leader and the algebraic connectivity of multi-agent system topology structure graph obtained by the optimization algorithm in [0,6]. Fig. 5(a) and (b) are the curves of algebraic connectivity evolution and virtual leader rate evolution trend. It can be seen that, at first, with the increase of virtual leader's speed, the generation connectivity increases to a certain extent; however, with the increase of virtual leader's speed, the connectivity begins to decrease, and virtual leaders tend to "separate" from common agent systems. Fig. 5(c) shows the difference between the position and speed of the UAV group center and the virtual navigator. It can be seen that the position and speed of the UAV group center tend to the position and speed of the virtual navigator during the flight. Then, the optimal speed of virtual leader is determined as 2.5 m/s. The experimental result also reflects that when the virtual leader speed is dynamically adjusted according to the neighbor state, on the basis of keeping the system connected, the speed of the virtual leader is continuously optimized through the fitness function constructed, so that the algebraic connectivity of the system gets a better value in the first and middle period, and the flocking speed of the multi-agent system is improved to a certain extent.

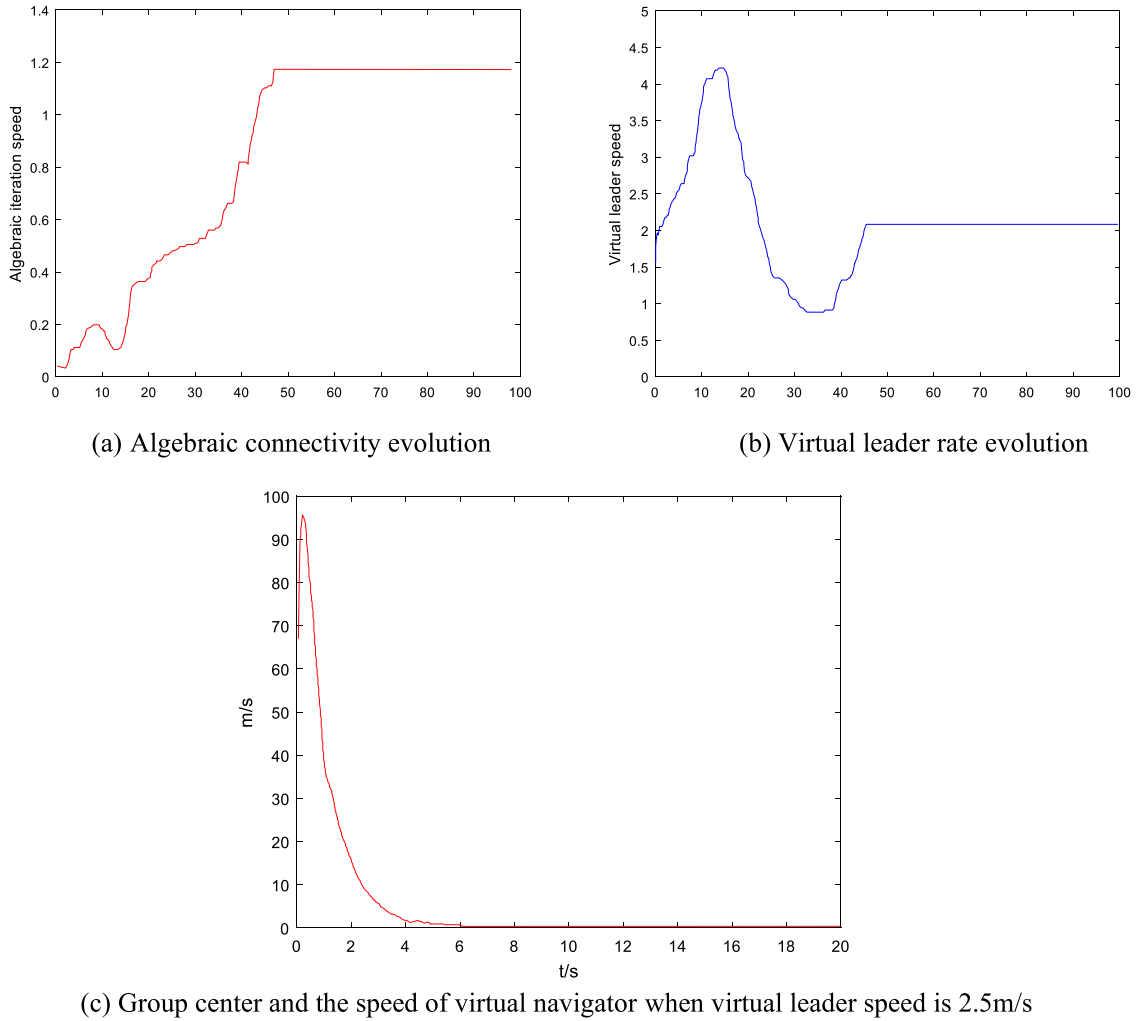


Fig. 5. Evolution graph of virtual leader's rate and algebraic connectivity.

4.2. Performance test

AODV/OLSR, which belongs to the on-demand routing protocol and the table driven routing protocol in the mobile ad hoc network, are used as the simulation protocols in the network layer. The impact of the formation control of this method on the data transmission performance is verified by the two indicators of network transmission delay and delay jitter. Taking the number of UAV nodes as a variable, two indexes that have the greatest impact on the communication of multiple UAVs are selected, namely, the average transmission delay of the network $\bar{D}$ and delay jitter $J(i)$ are taken as comparative items. The calculation of network average transmission experiment is shown as follows:

$$\bar{D} = \frac{1}{N} \sum_{i=1}^N D(i) \quad (34)$$

where $D(i) = RT(i) - ST(i)$ denotes the transmission delay of the i_{th} group; $RT(i)$ denotes the receiving time and $ST(i)$ denotes the sending time of the i_{th} group. The delay jitter is computed as $J(i) = D(i) - D(i - 1)$.

Fig. 6 shows the packet transmission delay under different UAV numbers and two different network protocols. It can be seen that, under the two routing protocols, the method in this paper has advantages. With the increase of the number of UAV nodes, the degree of delay rise of the method in this paper is smaller. When the number of UAV groups is $n = 40$, the transmission delay jitter under the two network protocols is described in Fig. 7. It can be concluded that in the whole

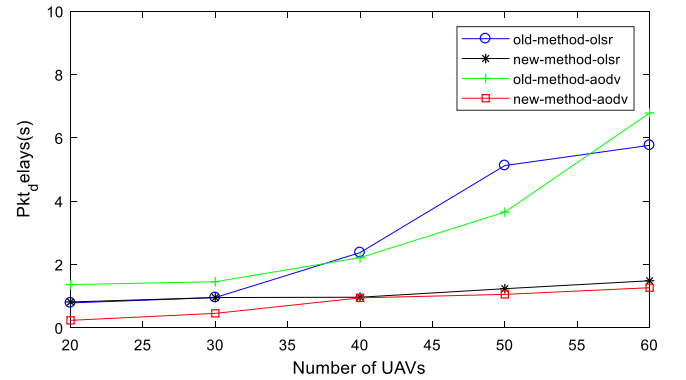


Fig. 6. Transmission delay comparison under two protocols.

process, the delay jitter of this method is not obvious, while in the time interval of 0–50 timesteps, the comparison method has obvious delay jitter. The average delay and delay jitter of the two indexes show that the method proposed in this paper makes the real-time communication between UAVs better and the information between UAV groups more timely.

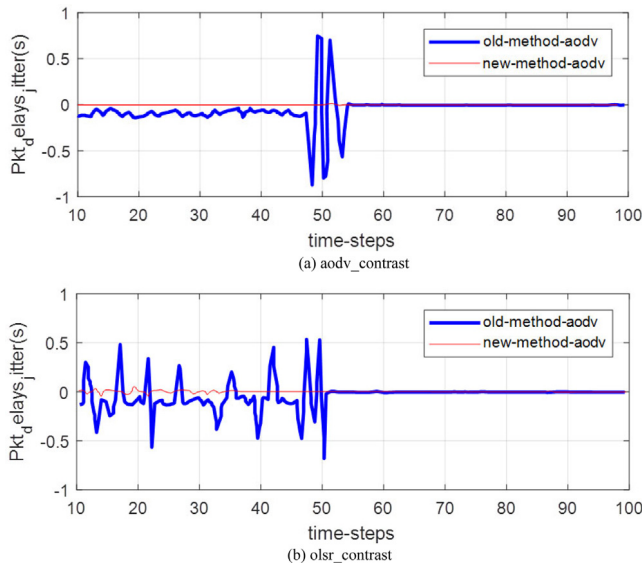


Fig. 7. Delay jitter comparison of algorithms.

5. Conclusion

With the development of computer technology and wireless communication technology, it is possible that a single robot system is replaced by a flocking system composed of multiple robots. The group UAV system has the advantages of strong environmental perception, high efficiency and strong robustness, which are difficult to grasp by a single robot, and the group UAV's security communication strategy control has become one of the important research contents in the field of multi robot cooperation. This article studies the security and consistency of group UAV Communication. Based on the existing group models, the improved flocking algorithm and the new target position moving function are used to enhance the stability of the formation and improve the effectiveness and convergence of the overall scheme algorithm.

However, most of the research in this article is on the optimal control of multi UAV network topology and formation. Although considering the communication range of different communication power for UAVs in different environmental positions, a position based power control model is also proposed. Due to the complexity of the network structure, the power control of mobile ad-hoc network involves multiple network level optimization methods. In the future work, more methods such as machine learning will be taken into account to study the power control algorithm.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Wei Liu: Data curation, Formal analysis, Funding acquisition, Writing - original draft. **Zhijun Gao:** Validation, Visualization, Investigation, Writing - review & editing.

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